

SAVE YOUR FRIDGE FOR FREE: FREEZE JUGS, NOT FOOD

When a winter storm kills the power, the cold outside can keep your food safe. The jug count, the swap schedule, and the food safety limits.

Fill plastic jugs with tap water, set them outside overnight, and put the ice in your dead fridge. It costs nothing if you own a thermometer that reads in the 30s. In February 2021, Winter Storm Uri cut power to about 4.5 million customer accounts in Texas, some for days. The cold that was ruining the pipes could have saved the fridge.

THE CLOCK

About 4 hours

A fridge holds a safe temperature with the door shut, per the Department of Agriculture.

About 48 hours

A full freezer. A half full one, about 24.

40°F, 2 hours

Perishables above 40 for more than 2 hours get thrown out: meat, milk, eggs, leftovers, soft cheese.

That 4 hours assumes a warm kitchen. Check the thermometer first: a kitchen already in the 40s gives you a lot longer. A house still near 65 means the clock is real.

BEFORE THE STORM

- ☐ Fill plastic bottles and jugs and freeze them in the freezer while it still runs. Fill every empty space: a full freezer holds about twice as long as a half full one.
- ☐ Turn the fridge and freezer to their coldest settings a day ahead.
- ☐ Put a thermometer in each one now, while you can still buy one.
- ☐ On a well? The pump dies with the power. Fill your jugs before the storm hits, or use snow.

THE FIRST DAY

- ☐ Keep the doors shut. Every opening spills cold air.
- ☐ Small bottles first: water bottles and 20 ounce soda bottles freeze in a few hours on a cold night. A gallon jug takes roughly a day on a night in the teens.
- ☐ Clean snow or ice? Pack it into zip top bags, seal them, and set them in a bowl for the meltwater.
- ☐ **Plastic only.** Water grows about 9 percent when it freezes, and glass can crack. Fill to the shoulder, leave about a tenth empty.



Jugs on a board, off the ground, under open sky on a freezing night.

SHOPPING LIST

- ☐ **Plastic jugs and bottles:** milk jugs, water bottles, soda bottles. Free if you save them.
- ☐ **Appliance thermometer:** 6 to 9 dollars. A digital instant read, or a weather thermometer with its probe in the fridge, works too.
- ☐ **Zip top bags** and a bowl, for snow.
- ☐ **A real cooler** (not cardboard), two scrap boards, a strap or a cinder block, a blanket or tarp.
- ☐ **Sealed containers** for anything that goes outdoors.

WHY NOT THE PORCH?

- ☐ The Department of Agriculture says don't put the food itself outside: temperatures swing, and animals and dirt get to it.
- ☐ Eggs on a 12 degree porch freeze by morning, and a frozen egg with a cracked shell gets thrown away. Lettuce, celery, and cucumbers turn to mush.
- ☐ Sun on a dark box adds a lot of heat. A clear, calm night runs colder than the air, which is why you scrape frost at 36 degrees.
- ☐ Their better move, in the next sentence: fill buckets, milk jugs, or cans with water, freeze them outside, and use the ice.

WHY ICE WORKS

About 144 BTU To melt 1 pound of ice. Warming that pound of water 1 degree takes just 1.

About 1,200 BTU Absorbed melting one gallon of ice (a gallon is about 8 1/3 pounds).

Close to 5,000 BTU of cooling from four gallon jugs in a fridge.

32°F Where a jug stays while it's part ice. It can't warm until the last ice melts.

About 1 lb The air in an empty fridge. It holds almost no cold, so frozen mass does the work.

Milk, meat, and eggs freeze a few degrees below 32. Milk freezes around 31. Nothing holds the fridge in a safe band on its own, so check the thermometer and keep delicate foods off the ice.

HOW MANY JUGS

10° gap Fridge at 35 in a 45 degree kitchen. Heat comes in slowly.

30° gap Fridge at 35 in a 65 degree house: heat comes in about 3 times faster.

About 1 gallon Of ice a day in a cold kitchen, as a rough estimate.

3+ gallons A day in a warmer house, plus every door opening.

Start with 6 gallon jugs. Swap 2 or 3 a day. Let the thermometer tell you if you need more.



Biggest jugs on the top shelf, food packed in the middle, jugs around it, thermometer mid shelf.

THE DAILY ROUTINE

- 1 Night:** set the jugs outside. Open sky and a little wind freeze them fastest. Off the ground on a board or a step, out of reach of dogs.
- 2 Morning:** bring the frozen ones in. Biggest jugs on the top shelf, because cold air sinks. Food packed together in the middle, other jugs around it.
- 3 Day:** spare frozen jugs go in shade, like a cooler on the north side of the house.
- 4** Thermometer on the middle shelf. Read it every time you open the door.
- 5** Climbing toward 40? Add ice. A fully melted jug goes back out to refreeze. The water never gets used up.

A cold reading can't prove the food stayed under 40 the whole time, so check often.

WHAT TO SAVE FIRST

Fine at room temperature	Coldest spot, closest to the ice
Hard cheese, butter, peanut butter, jelly, ketchup, mustard, bread, whole raw fruit	Raw meat, poultry, fish, milk, soft cheese, eggs, cooked leftovers, anything with mayonnaise

Can't keep it all cold? Eat or cook the most perishable things first.

THE FREEZER

- ☐ Ice holds at 32, and meat thaws a few degrees below that. Jugs keep freezer food cold and safe, but not hard frozen for long.
- ☐ Nights in the teens and days below freezing? An outdoor cooler can be your freezer for frozen meat. Eggs, milk, and produce can't go, because they'll freeze.
- ☐ One box for frozen food, one for fridge food, a thermometer in each.

THE OUTDOOR COOLER

- ☐ A real cooler, on the **north side** of the house, in full shade, against the wall.
- ☐ Off the ground on a couple of scrap boards.
- ☐ At night, cover the lid with a blanket or tarp so it isn't losing heat to the open sky.
- ☐ Every item in a sealed container. Raccoons, rats, and dogs find a cooler fast. Strap the lid or set a cinder block on it.
- ☐ Read the thermometer every morning and evening. **Under 32?** That box is a freezer now, so fridge food comes out.

About R 10

Foam walls on paper, about 13 square feet of surface on a midsize cooler.

About 30 BTU/hr

Leak on a 10 degree night, 22 degrees colder than the 32 you want inside.

Over a day

One gallon jug of liquid water, giving back about 1,200 BTU as it freezes, can hold it near 32. Best case: real coolers leak at the lid.

FOOD SAFETY RULES. NOT OPTIONAL.

- ☐ Fridge food above **40°F for more than 2 hours**, or for a time you don't know: throw it out.
- ☐ Freezer food with ice crystals, or reading 40 or below, can be refrozen safely. It may lose texture, but it's safe.
- ☐ Don't know how long it sat above 40? Throw it out.
- ☐ **Never taste food to decide.** The bacteria that make you sick don't change the smell or the look.
- ☐ Don't put food straight on the porch. Only jugs go outside bare.
- ☐ Plastic only for freezing. Glass can crack.

WHEN THE POWER COMES BACK

- 1 Check the thermometers before anything else.
- 2 Refreeze what qualifies, toss what doesn't, using the rules above.
- 3 Empty the meltwater and wipe the shelves.
- 4 Put the jugs back outside. Next storm, they're already waiting.

VERSUS A GENERATOR

- ☐ A fridge uses about as much power in a day as a 100 watt bulb left on for 10 to 20 hours. Shift running still burns a few gallons of gas a day, around 10 to 15 dollars, plus the generator.
- ☐ Generator outside only, at least **20 feet** from the house, exhaust away from doors, windows, and vents. A battery powered carbon monoxide alarm outside every sleeping area.
- ☐ During Uri, generator exhaust was tied to at least 10 deaths. Texas counted 246 deaths across 77 counties, most from hypothermia. Frozen jugs make no exhaust at all.

SUMMER BLACKOUT?

No free cold outside, so the jugs must be frozen before the power dies. Watch our video on keeping your fridge cold for 3 days when the power dies.



North side, shade, off the snow, lid weighted and covered. The raccoon is why.

Watch the full video: youtube.com/@SketchySurvival101

More free sheets: sketchysurvival101.com/free

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